

Language as Boundary Condition: From Intuition to Theorem

A Genealogy of the Unformalized and the Formal in Structural Philosophy

Claudio Bresciano

Abstract

The thesis that language functions as a boundary condition on thought —not merely as its vehicle but as its geometric limit— has appeared in various forms throughout the history of philosophy. Wittgenstein saw language as form of life; Heidegger as the house of Being; Derrida as the play of *différance*; Foucault as the archive of discourse. Yet none of these thinkers formalized the intuition as a mathematical structure. None demonstrated that the "something" philosophy perennially seeks is the necessary collapse of a space of possibilities into a realized trajectory. This paper traces the genealogy of the intuition, identifies the precise point where each predecessor stopped, and presents the Theory of Axiomatic Necessity (TNA) as the first formalization of language as boundary condition. The contribution is not the intuition but the theorem: the diagonal argument proving that a language of pure relations without relata is structurally impossible, and the collapse operator Ψ showing that the "something" is not a discovery but a necessary condition of interpretive closure.

1. The Intuition: A Shared Recognition Across Traditions

The history of philosophy records a recurring insight: language is not neutral. It conditions, constrains, and constitutes what can be thought. This insight appears in multiple registers:

THINKER	REGISTER	FORMULATION	CENTURY
Wittgenstein	Pragmatic	"The limits of my language are the limits of my world"	20th
Heidegger	Ontological	"Language is the house of Being"	20th
Derrida	Structural	"There is nothing outside the text" / <i>différance</i>	20th

THINKER	REGISTER	FORMULATION	CENTURY
Foucault	Historical	Episteme as condition of discourse	20th
Sapir-Whorf	Empirical	Linguistic relativity hypothesis	20th
Chomsky	Biological	Universal Grammar as innate competence	20th

The intuition is shared. The formalization is absent.

2. Wittgenstein: The Form of Life as Unformalized Boundary

In the *Tractatus Logico-Philosophicus* (1921), Wittgenstein states:

5.6 "Die Grenzen meiner Sprache bedeuten die Grenzen meiner Welt."
(The limits of my language mean the limits of my world).

This is a boundary condition in natural language. But Wittgenstein does not formalize it. He does not ask: *What is the structure of a system that requires an external boundary to achieve closure?*

Instead, he moves to the *Philosophical Investigations* (1953) and replaces the boundary with practice:

"For a large class of cases —though not for all— in which we employ the word 'meaning' it can be defined thus: the meaning of a word is its use in the language." (§43)

The boundary becomes internal to the language-game. There is no N_1 ; there are only language-games (N_0 systems) that overlap and compete. Wittgenstein's later work dissolves the boundary into pluralism.

- **Where Wittgenstein stopped:** He saw the boundary but dissolved it into practices. He did not prove that a pure system of practices (pure N_0) is structurally impossible.
-

3. Heidegger: The House of Being as Ontological Boundary

In the *Letter on Humanism* (1947), Heidegger writes:

"Die Sprache ist das Haus des Seins."
(Language is the house of Being).

This is stronger than Wittgenstein. Heidegger does not say language is a tool; he says it is the dwelling of Being. The human being does not speak language; language speaks the human being.

The distinction between Being (*Sein*) and beings (*Seiendes*) maps partially onto TNA:

HEIDEGGER	TNA
Being (<i>Sein</i>)	N_1 (condition of possibility)
beings (<i>Seiendes</i>)	N_0 (operational entities)
The ontological difference	The N_0/N_1 distinction

But Heidegger rejects formalization. In *Introduction to Metaphysics* (1935), he says: "*The question of Being is not a question of logic.*" For Heidegger, the boundary is hermeneutic, not mathematical. It is disclosed through poetry, not proof. The ontological distinction is not a theorem; it is an event (*Ereignis*).

- **Where Heidegger stopped:** He saw the boundary as ontological necessity but refused to formalize it. He chose poetic disclosure over axiomatic demonstration.

4. Derrida: Différance as Infinite Regress

Derrida's *différance* (1967) is the most radical version of the boundary intuition. Derrida argues that:

- Meaning is always deferred (*différance* = difference + deferral).
- There is no presence; only traces.
- The signifier never reaches the signified.
- Every "something" is a supplement that displaces another supplement.

This appears to be the regress argument of TNA:

$$R(1) \leftarrow R(2) \leftarrow R(3) \leftarrow \dots$$

But Derrida celebrates the regress. He does not seek closure. For Derrida, the "something" is not a necessary collapse but an impossible dream. The trace never becomes presence; the supplement never becomes origin.

- **Where Derrida stopped:** He saw the regress but denied the collapse. He chose dissemination (Σ without σ^*) over interpretation:

$$\Psi(\Sigma) = \sigma^*$$

5. Foucault: The Episteme as Historical Boundary

In *The Order of Things* (1966), Foucault analyzes the *episteme* —the historical *a priori* that determines what can be said in a given epoch. The episteme is a boundary condition on discourse:

"In any given culture and at any given moment, there is always only one episteme that defines the conditions of possibility of all knowledge."

But Foucault's episteme is historical, not universal. It changes:

- Renaissance episteme (resemblance)
- Classical episteme (representation)
- Modern episteme (man as object and subject)

There is no theorem of epistemic necessity. There is only archaeology —the description of historical shifts.

- **Where Foucault stopped:** He saw the boundary as historical contingency, not structural necessity. He chose genealogy over geometry.
-

6. Chomsky: Universal Grammar as Biological Boundary

Chomsky's Universal Grammar (UG) is the most formal predecessor. Chomsky argues that:

- Language is not learned; it is innate.
- The child is born with a Language Acquisition Device (LAD).
- UG is a biological structure that constrains possible grammars.

This is a boundary condition in the literal sense: UG is the set of possible human languages, and particular languages are selections within that space. But Chomsky's boundary is empirical, not axiomatic. It is discovered by linguistic analysis, not proved by mathematical necessity. The *Poverty of the Stimulus* argument is inductive, not deductive.

- **Where Chomsky stopped:** He formalized the boundary (transformational grammar) but did not prove its necessity. He chose biology over metamathematics.

7. The Gap: What None of Them Did

THINKER	SAW THE BOUNDARY?	FORMALIZED IT?	PROVED NECESSITY?	IDENTIFIED COLLAPSE?
Wittgenstein	Yes	No	No	No (dissolved into practices)
Heidegger	Yes	No	No	No (poetic disclosure)
Derrida	Yes	No	No	No (denied collapse)
Foucault	Yes	No	No	No (historical contingency)
Chomsky	Yes	Partially	No	No (biological, not axiomatic)

The gap is precise: none formalized the boundary as a mathematical condition of interpretive closure; none proved that a pure system of relations without relata is impossible; none identified the "something" as the necessary collapse:

$$\Psi(\Sigma) = \sigma^*$$

8. TNA: The Formalization

8.1 The Framework

TNA distinguishes two domains:

DOMAIN	SYMBOL	ROLE
Operational	N_0	Generates trajectories, states, processes
Admissibility	N_1	Selects which trajectories are realizable

8.2 The Diagonal Argument: Formal Version

8.2.1 Preliminaries

Let L be a first-order language with:

- **Signature:** $\Sigma_L = \langle \mathcal{R}, \mathcal{F}, \mathcal{C} \rangle$ where:
 - $\mathcal{R} = \{R_i(n_i) : i \in I\}$ is a countable set of relation symbols with arities $n_i \geq 1$.
 - $\mathcal{F}$ is a set of function symbols (possibly empty).
 - $\mathcal{C}$ is a set of constant symbols (possibly empty).
- **Domain:** D of interpretation.
- **Semantics:** Standard Tarskian semantics $\mathcal{I}$.

Definition 8.2.1 (Pure-Relation System):

L is a *pure-relation system* if:

1. $\mathcal{F} = \emptyset$ and $\mathcal{C} = \emptyset$ (no functions, no constants).
2. Every $R_i \in \mathcal{R}$ is interpreted as a relation $R_i^{\mathcal{I}} \subseteq D^{n_i}$.
3. There are no distinguished elements in D (no "ur-elements" or "substances").

For the purposes of this diagonal argument, we restrict to the subset $\mathcal{R}_1 \subseteq \mathcal{R}$ of unary relations (predicates) — those with arity $n_i = 1$.

Justification: The standard diagonal arguments (Russell's paradox, Gödel's lemma, Tarski's theorem) operate on unary predicates or formulas with one free variable. This restriction eliminates any technical objection regarding arities while preserving full generality, since any n -ary relation can be coded as a unary predicate over n -tuples.

Definition 8.2.2 (Interpretive Closure):

L has *interpretive closure* if for every well-formed formula ϕ in L , the truth value $\mathcal{I}(\phi) \in \{\top, \perp\}$ is determined by the interpretation of its constituent relation symbols.

Definition 8.2.3 (Admissibility Structure):

An *admissibility structure* for L is a set $\mathcal{A} \subseteq \mathcal{P}(D)$ of distinguished subsets of D that serve as the range of quantifiers and the domain of relational arguments.

8.2.2 Gödel Numbering

Since $\mathcal{R}_1$ is countable, fix an enumeration $\{P_1, P_2, P_3, \dots\}$ where each P_i is a unary predicate. Define an injective Gödel numbering:

$$\ulcorner \cdot \urcorner : \mathcal{R}_1 \rightarrow \mathbb{N}^+$$

For each P_i , its self-application code is the number $\ulcorner P_i \urcorner$.

8.2.3 The Construction

Assume L is a pure-relation system with interpretive closure. Let $\mathcal{R}_{\text{def}} \subseteq \mathcal{R}_1$ be the set of definable predicates in L : those P_i for which there exists a formula $\phi_i(x)$ with:

$$\mathcal{I} \models P_i(d) \iff \mathcal{I} \models \phi_i(d)$$

Lemma 8.2.1: *If L has interpretive closure, then $\mathcal{R}_{\text{def}}$ is countable.*

Proof: Immediate from countability of the set of formulas in a first-order language with a countable signature. ■

Enumerate $\mathcal{R}_{\text{def}} = \{P_1, P_2, P_3, \dots\}$ with Gödel numbers $\ulcorner P_i \urcorner = i$.

Definition 8.2.4 (The Diagonal Predicate):

Define $D(x)$ as:

$$\text{Kish} D(n) \iff \exists P_i \in \mathcal{R}_{\text{def}} : \ulcorner P_i \urcorner = n \wedge \neg P_i(n)$$

where $P_i(n)$ means $\mathcal{I} \models P_i(\bar{n})$ with $\bar{n}$ being the formal numeral for n .

8.2.4 The Theorem

Theorem 8.2.1 (Impossibility of Pure-Relation Closure):

No pure-relation system L with interpretive closure can contain its own admissibility structure.

Proof:

Suppose, for contradiction, that L contains its own admissibility structure. Then every definable predicate in L is recognized by L 's interpretive apparatus. Consider the diagonal predicate $D(x)$ defined above.

- **Step 1:** $D(x)$ is definable in L because:
 - The set $\mathcal{R}_{\text{def}}$ is definable (by Lemma 8.2.1).
 - The Gödel numbering is fixed.
 - Negation and existential quantification are fundamental operators in L .
 - **Step 2:** Since $D(x)$ is definable, there exists some $P_k \in \mathcal{R}_{\text{def}}$ such that:

$$\mathcal{I} \models P_k(n) \iff \mathcal{I} \models D(n)$$
 - **Step 3:** Instantiate at $n = \ulcorner P_k \urcorner = k$:

$$\mathcal{I} \models P_k(k) \iff \mathcal{I} \models D(k) \iff \mathcal{I} \models \neg P_k(k)$$
 - **Step 4:** This yields the structural contradiction:

$$\mathcal{I} \models P_k(k) \iff \mathcal{I} \models \neg P_k(k)$$
 which is a contradiction in classical bivalent logic.
 - **Step 5:** Therefore, the assumption that L contains its own admissibility structure is false.
-

8.2.5 The Corollary: Necessity of Non-Relational Terminus

Corollary 8.2.1: *Every system L with interpretive closure requires an extension $L^+ \supset L$ such that:*

1. L^+ contains a distinguished element $a^* \in D^+$ (a "substantive anchor").*
2. L^+ contains a predicate P^* that is not definable in L .*
3. P^* serves as the admissibility structure for L .

Proof:

By Theorem 8.2.1, L cannot contain its own admissibility structure. Therefore, the admissibility structure must be external to L . Let L^+ extend L with:

- A new constant symbol c^* interpreted as $a^* \in D^+$.
- A new predicate symbol P^* interpreted as a distinguished subset of D^+ .

The extension L^+ provides the "something" —the non-relational terminus— that L cannot generate internally.

To verify that P^* is not definable in L : suppose it were. Then P^* would be in $\mathcal{R}_{\text{def}}$, and the diagonal argument would apply to L^+ as well, generating a new contradiction. Therefore, P^* must be primitive in L^+ .

8.2.6 Connection to TNA

In TNA terminology:

- L is N_0 (operational domain).
- L^+ is $N_0 \cup N_1$ (operational + admissibility domain).
- P^* is the selector that collapses Σ into σ^* .
- The extension $L \subset L^+$ is the structural operation:

$$\Psi : \Sigma \rightarrow \sigma^*$$

The "something" that philosophy seeks is precisely this P^* : the primitive predicate that every operational system must import to achieve interpretive closure.

8.2.7 Distinction from Tarski

Remark: While Tarski's theorem demonstrates that Truth (a semantic property) cannot be defined within L , Theorem 8.2.1 demonstrates that Admissibility (a structural/selector property) cannot be derived within L .

- **Tarski closes the semantic hierarchy:** no language can contain its own truth predicate.
- **TNA closes the realizability hierarchy:** no operational system can contain its own selector.

The theorems are isomorphic in structure but distinct in object. Tarski governs meaning; TNA governs possibility.

8.3 Comparison with Established Results

RESULT	DOMAIN	STRUCTURE	CONCLUSION
Cantor's Theorem	Sets	Diagonalization of power set	$\ A\ < \ \mathcal{P}(A)\ $
Russell's Paradox	Naive set theory	$R = \{x : x \notin x\}$	Inconsistency of unrestricted comprehension

RESULT	DOMAIN	STRUCTURE	CONCLUSION
Gödel's First Incompleteness	Formal arithmetic	Gödel sentence $G \iff \neg \text{Prov}(\ulcorner G \urcorner)$	$\not\models G$ and $\not\models \neg G$
Tarski's Undefinability	Truth in L	Truth predicate $\text{Tr}(x)$	Tr not definable in L
TNA Diagonal	Pure-relation systems	Diagonal predicate $D(x)$	Admissibility structure not internal to L

9. The Difference in One Sentence

The predecessors had the intuition that language bounds thought; TNA has the theorem that the bound is a necessary collapse from multiplicity to singularity, and that the "something" is the signature of this collapse.

10. Conclusion: From Poetry to Proof

The history of philosophy is a history of approximations to the same structural insight:

- Wittgenstein approached it through grammar.
- Heidegger through poetry.
- Derrida through deconstruction.
- Foucault through archaeology.
- Chomsky through biology.

TNA approaches it through geometry. Not the geometry of space, but the geometry of possibility —the structure of spaces that require external conditions to achieve closure.

The "something" is not a mystery. It is the fulcrum of the lever. It is the boundary condition of the equation. It is the collapse of the wave function. It is the system prompt of the LLM. It is the axiom of the formal system.

It is obvious. But, as Archimedes showed, the obvious is not always seen.

References

- Chomsky, N. (1965). *Aspects of the Theory of Syntax*. MIT Press.
- Derrida, J. (1967). *De la grammatologie*. Minuit.
- Foucault, M. (1966). *Les mots et les choses*. Gallimard.
- Heidegger, M. (1927). *Sein und Zeit*. Niemeyer.
- Heidegger, M. (1947). "Brief über den Humanismus." In *Wegmarken*. Klostermann.
- Wittgenstein, L. (1921). *Tractatus Logico-Philosophicus*. Routledge & Kegan Paul.
- Wittgenstein, L. (1953). *Philosophical Investigations*. Blackwell.
- Bresciano, C. (2025). *Core Structure of the Theory of Axiomatic Necessity*. Zenodo. DOI: 10.5281/zenodo.18098558
- Bresciano, C. (2026). *Language as a Boundary Condition of Philosophy: Why We Always Search for "Something"*. Zenodo. DOI: 10.5281/zenodo.20546730
- Bresciano, C. (2026). *Boundary Conditions as Structural Selectors: A Formal Analysis of N_0 , Operational Constraints, and N_1 in Self-Optimizing Systems*. Zenodo. DOI: 10.5281/zenodo.20479474
- Bresciano, C. (2026). *Castello Aereo Non Stat: TNA as the Archimedean Principle of Foundation*. Zenodo. DOI: 10.5281/zenodo.20597938

Appendix: The Predecessors and TNA in Parallel

PROBLEM	WITTGENSTEIN	HEIDEGGER	DERRIDA	TNA
What bounds thought?	Language-games	Being	Text / <i>différance</i>	N_1 as admissibility structure
What is the "something"?	Use	Being	Trace / supplement	Collapse $\Psi(\Sigma) = \sigma^*$
How is the bound known?	Practice	Poetry	Deconstruction	Diagonal argument
Is closure possible?	Yes (local)	Yes (event)	No (infinite)	Yes (necessary, by theorem)
Is the bound formalizable?	No	No	No	Yes ($N_0/N_1, \Psi, \aleph$)